

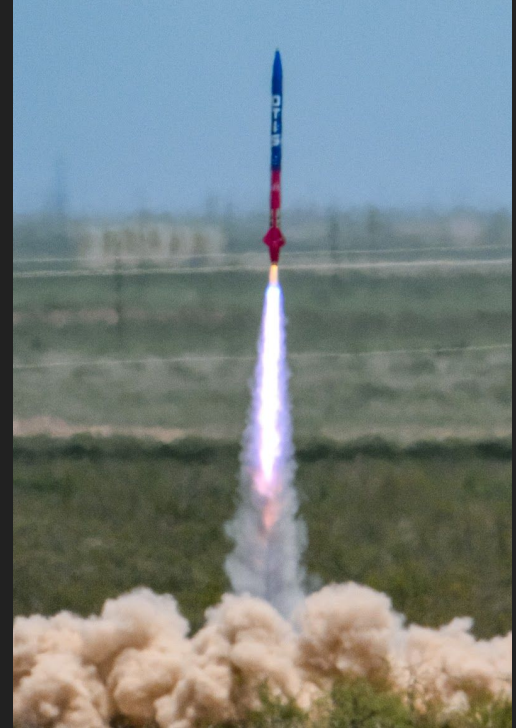
Classifying Phases in High Powered Rocketry

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Problem

High Powered Rocketry

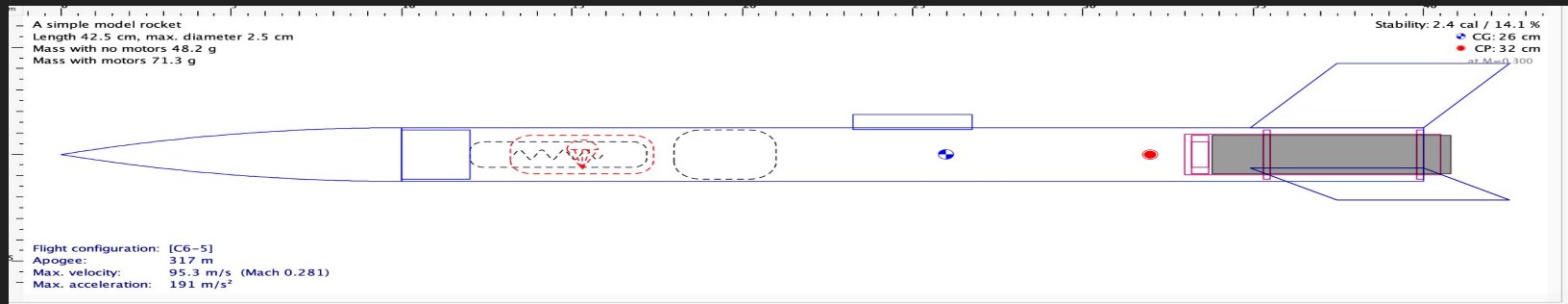
- Variety of rockets that can fly up to ~30k ft AGL
- Several phases
 - Pad
 - Boost
 - Coast
 - Apogee
 - Parachute Events
 - Landing
- Rockets come in different shapes and sizes, and can use a range of rocket motors
- Detecting these events quickly is important considering how fast a rocket travels



Data

Data Generator

- Utilized rocket configurations from RIT's rocketry group.
 - Length / Weight variation -
 - Velocity / Acceleration variation -
- A script was used to generate data based on this information using OpenRocket, a rocket launch simulator
- Injected noise into this data to more accurately demonstrate a rocket launch



Data Format

Data has 5 features

- Time
- Air temperature
- Air Pressure
- Acceleration on X Y plane
- Acceleration on Z plane

Data has 1 target

- Flight Phase

1	TIME	AIR_TEMPERATURE	AIR_PRESSURE	ACCELERATION_XY	ACCELERATION_Z	flight_phase
2	0.01	288.31424045027900	101325.4299140850	0.0025148620967198000	50.89615635140840	1
3	0.02	288.66129258578100	101315.42553538100	-0.008975414878401990	110.30488285444000	1
4	0.03	288.30694565671200	101338.19365148500	0.011056500799770100	115.90900323684000	1
5	0.04	287.84101611543100	101337.6136833380	-0.0071671182385739500	112.74864246928400	1
6	0.05	289.14770681129800	101343.18302276900	-0.0032718947025004300	111.10450126115700	1
7	0.06	287.908928974272	101326.66038181000	-0.004173478825218910	111.00791074723700	1
8	0.07	288.1814977965620	101323.92677126900	0.025037265263112300	110.89133533127500	1
9	0.08	288.3118264661610	101320.37004104300	0.007126894022697990	110.78082291990800	1
10	0.09	288.76290316709	101316.90448997300	-0.011138333828005100	110.65539467783700	1
11	0.09999999999999999	289.0480056990990	101305.09283247900	0.01527259036425650	110.54324087745100	1
12	0.11000000000000000	287.96857912129800	101318.15118778100	0.00452795441539762	110.45122269891800	1
13	0.12000000000000000	288.4235410051850	101317.6639472560	-0.007080119362343280	110.31491165059800	1
14	0.13000000000000000	287.98556432403400	101303.16608389300	-0.02233297990326630	110.2023616142440	1
15	0.14000000000000000	288.1661919248460	101326.0233809440	-0.005643240563867310	110.08648479499500	1
16	0.15	287.94505380369300	101302.16613352400	-0.013665899900506600	109.9573875555860	1
17	0.16	288.23277276005100	101298.5402412630	-0.0050307155519720900	109.84484198001200	1
18	0.17	288.45496211261900	101314.26572301200	0.004910379456245370	109.73410425015200	1
19	0.18	289.04387500498700	101298.78825280200	0.01043725686578300	109.5975098062230	1
20	0.19	288.2175769929350	101288.850787685	0.0017717128463160500	109.47440850023200	1
21	0.2	288.89644556765400	101303.23003946400	0.005503031287219570	109.34642159515500	1
22	0.21	288.560696592073	101306.61920168800	-0.0007937131660039180	109.17194296561800	1
23	0.22	287.0368766293870	101301.00833826200	-0.002204060560899720	108.90961157683800	1
24	0.23	287.8796471064760	101292.76942461800	0.007771591366628890	108.6480413463630	1
25	0.24	288.22389735880800	101290.00400988100	-0.00425262844861912	108.40776526726700	1
26	0.255	288.50396336637800	101291.95870603600	0.020528191404733600	108.08236452999600	1
27	0.27750000000000000	288.10820879505400	101281.65405861200	0.011325275496011300	107.61847625944700	1
28	0.31125000000000000	287.3375248500530	101266.7507843880	0.005105240163812410	106.9104471257720	1
29	0.36125	289.4804629539650	101244.27189116500	0.011772897156829200	105.82647635791200	1

Models

Logistic Regression

- Map extracted temporal features to class probabilities using a linear decision boundary
- First version flattened a window into a 400 element vector
 - Window of shape (4, 100) becomes a 400 element vector
 - Achieves .75 accuracy
 - Achieves .82 weighted F1 score
- Newer version extracted data into 24 features
 - Mean, STD, min, max, slope, delta across 6 channels
 - Achieves .95 accuracy
 - Achieves .95 weighted F1 score

KNN with Dynamic Time Warping (DTW)

- KNN regression classifier using a K factor for 5.
- Uniform Weights: all data is weighted equally
- Metric: DTW used to calculate distance
 - DTW is an algorithm to compare non-linear sequences of data.
- KNN decide algorithm to calculate nearest neighbor.
- Train / Test split:

Achieved results

- Test / Train Accuracy: 0.995 / 0.993
- Weighted F1: 0.98

Time Series Forest

- Ensemble of decision trees built on random time series intervals
- N Estimators: 200 trees trained in parallel
- Feature Extraction: mean, standard deviation, and slope computed per interval
- Multivariate input across 4 channels: acceleration, velocity, altitude, time
- Train / Test split: 80 / 20

Achieved results

- Test / Train Accuracy: 0.99 / 1.0
- Weighted F1: 0.9984 / 1.0
- Macro F1: 0.9823

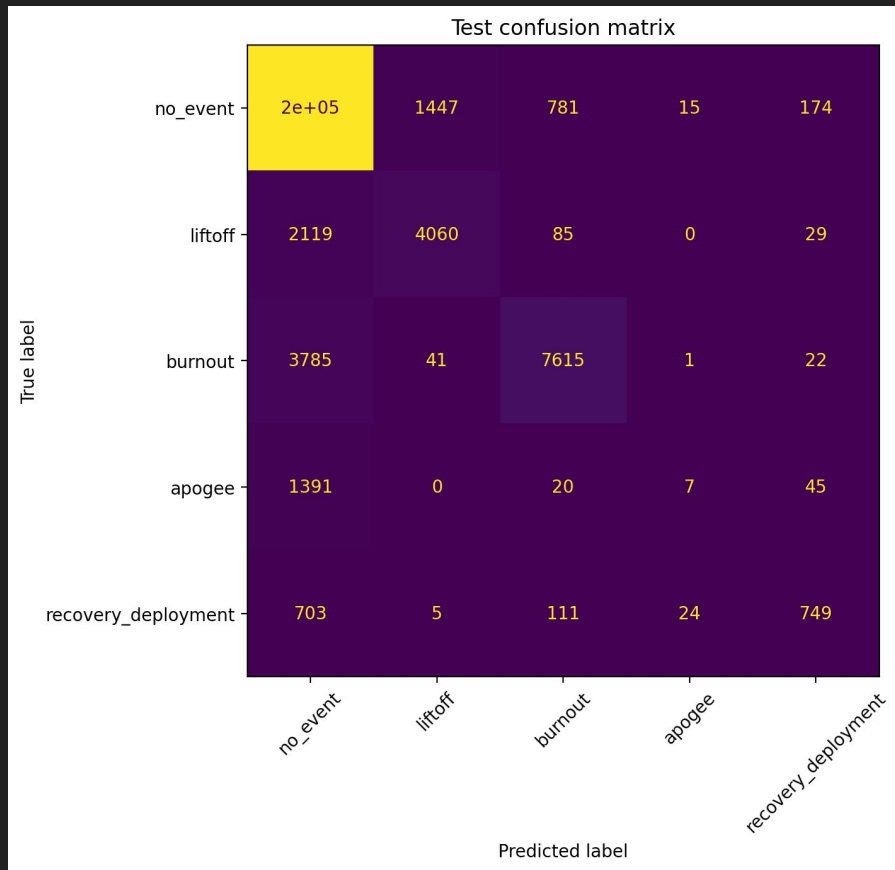
MINIROCKET

- Applies large set of random convolutions to time series
- Kernels produces two scalar features
 - Max value
 - Proportion of positive values (PPV) across the convolution output
 - Fixed and random making it faster to train compared to some aforementioned models
 - Random kernels are effective at capturing diverse temporal patterns without any domain knowledge
- Linear classifier on top makes it still interpretable at the feature level

Results

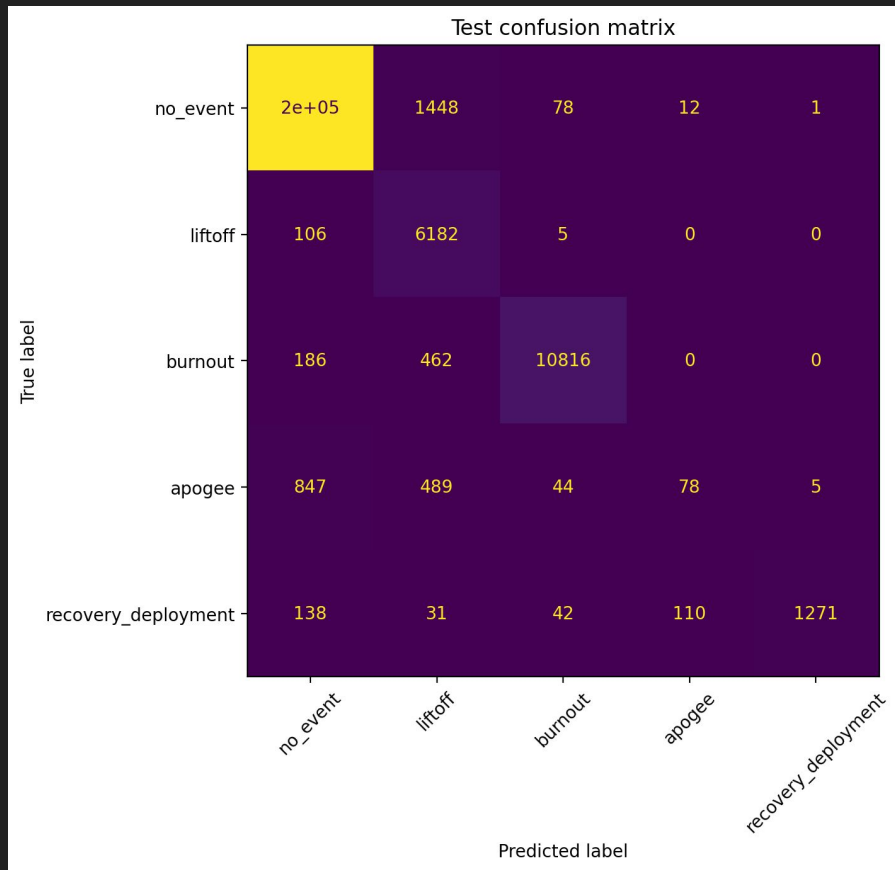
Logistic Regression

- Train
 - Accuracy: 0.95
 - F1: 0.61
 - Pad: 0.97
 - Liftoff: 0.69
 - Burnout: 0.76
 - Apogee: 0.01
 - Recovery: 0.57
 - Train Time: 7m 34s
- Test
 - Accuracy: 0.95
 - F1: 0.60
 - Pad: 0.97
 - Liftoff: 0.69
 - Burnout: 0.75
 - Apogee: 0.02
 - Recovery: 0.58
 - Inference Time: 680 ms



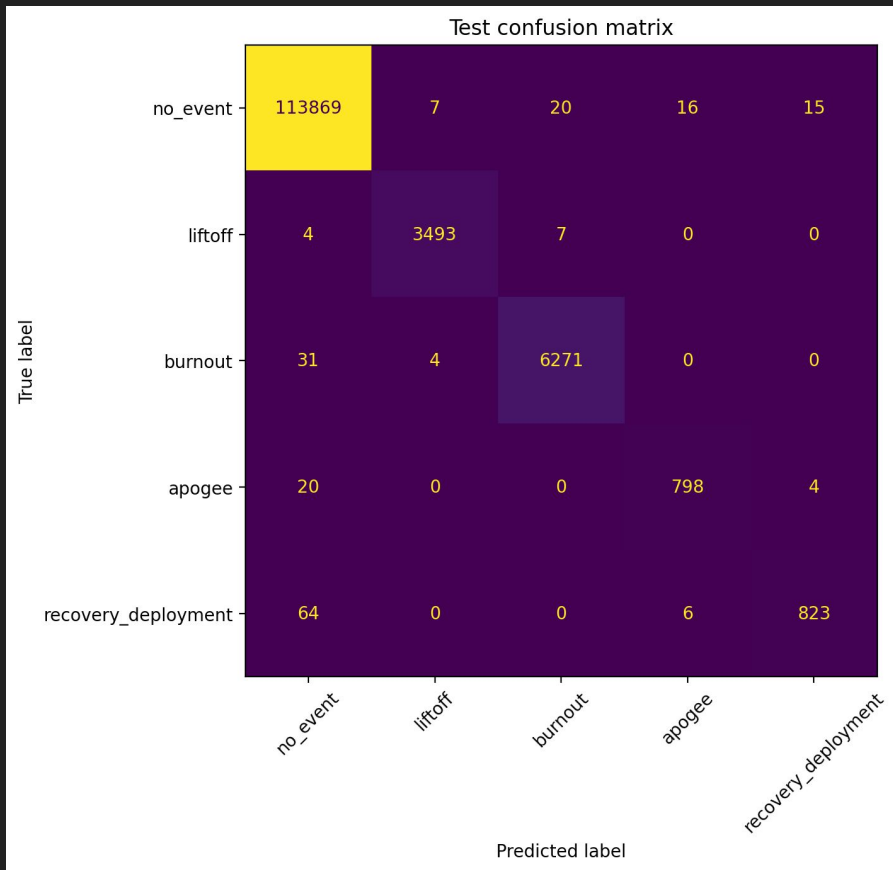
KNN w/ DTW

- Train
 - Accuracy: 0.98
 - F1: 0.77
 - Pad: 0.99
 - Liftoff: 0.84
 - Burnout: 0.97
 - Apogee: 0.13
 - Recovery: 0.90
 - Train Time: 59s
- Test
 - Accuracy: 0.98
 - F1: 0.75
 - Pad: 0.99
 - Liftoff: 0.83
 - Burnout: 0.96
 - Apogee: 0.09
 - Recovery: 0.89
 - Inference Time: 31m



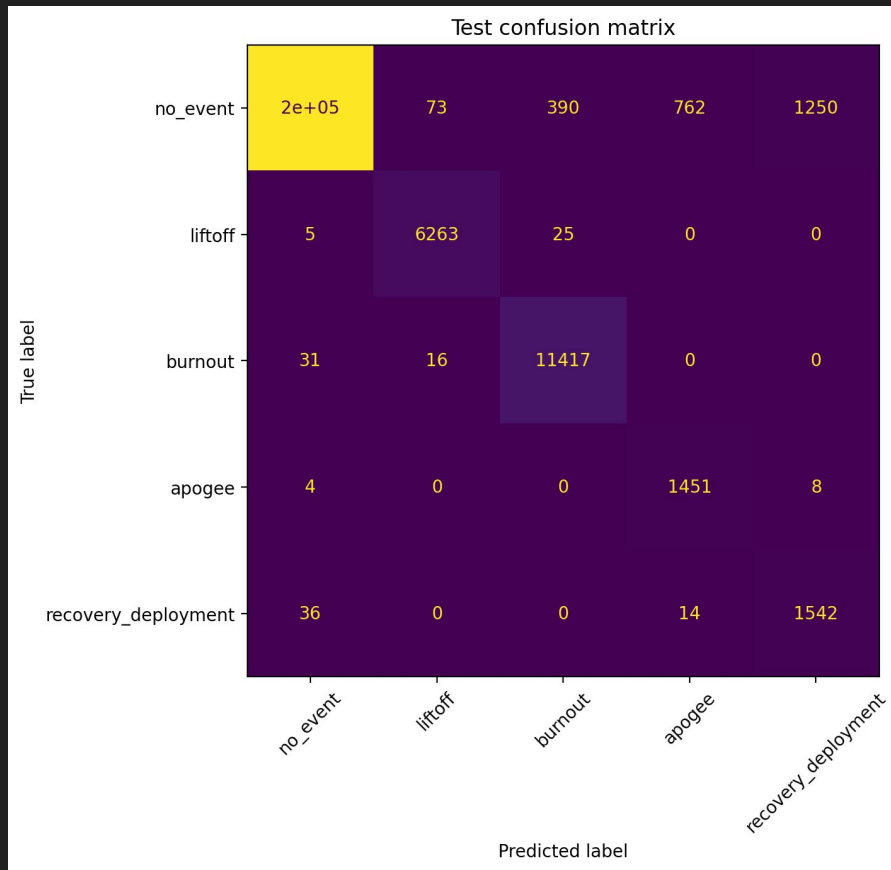
Time Series Forest

- Train
 - Accuracy: 1.0
 - F1: 1.0
 - Pad: 1.0
 - Liftoff: 1.0
 - Burnout: 1.0
 - Apogee: 1.0
 - Recovery: 1.0
 - Train Time: 2 hrs 10 minutes
- Test
 - Accuracy: 0.99
 - F1: 0.998
 - Pad: 0.9999
 - Liftoff: 0.9969
 - Burnout: 0.9951
 - Apogee: 0.9719
 - Recovery: 0.9488
 - Inference Time: 37 minutes



MINIROCKET

- Train
 - Accuracy: 0.99
 - F1: 0.90
 - Pad: 0.99
 - Liftoff: 0.99
 - Burnout: 0.98
 - Apogee: 0.80
 - Recovery: 0.72
 - Train Time: 2h 44m 6s
- Test
 - Accuracy: 0.99
 - F1: 0.90
 - Pad: 0.99
 - Liftoff: 0.99
 - Burnout: 0.98
 - Apogee: 0.80
 - Recovery: 0.71
 - Inference Time: 10s



Model	Train Acc	Test Acc	Train F1	Test F1	Train Time	Inference Time (per window)
Logistic Regression	0.95	0.95	0.61	0.60	7m 34s	0.003 ms
KNN w/ DTW	0.98	0.98	0.77	0.75	59s	8.25 ms
Time Series Forest	1.0	0.99	1.0	0.998	2h 10m	17.58 ms
MINIROCKET	0.99	0.99	0.90	0.90	2h 44m 6s	0.044 ms

Analysis

- Logistic Regression is the fastest at 0.003 ms/window
- MiniRocket provides the best accuracy-speed trade off at 0.044 ms/window
- KNN and TSF can both be real-time capable depending on constraints
- KNN's slow inference is because it brute-force searches across 900k training windows without compression
- Window level splits instead of flight level splits may have led to some data leakages leading to worse generalization on more unique, unseen rockets
 - Rockets with similar flight profiles had F1 scores ranging from 60%-90%

Questions?